Matteo Salis

Postdoctoral Researcher
AI for Weather Forecasting

Météo France: CNMR/GMAP/PREV

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Work Experience

03/2026 (current) **Postdoctoral Researcher (CDD)**, *Météo France, CNMR/GMAP/PREV team*, Toulouse, FR.
I'm working on the development and fine-tuning of a foundational model for high-resolution medium-range weather forecasting within the Weather Generator Horizon Project.

Education

2022–2025 **PhD in Modeling and Data Science**, *University of Turin & Université Savoie Mont Blanc - UNITA Cotutelle agreement*, Torino, IT / Annecy, FR.
Thesis: *Deep Learning models for spatio-temporal environmental phenomena and applications on groundwater resources.*
Supervisors: Rosa Meo, Stefano Ferraris, Abdourrahmane M. Atto.

2020–2022 **MSc Environmental Economics (LM56)**, *University of Turin*, Italy
110/110 cum laude and right of publication. Thesis: *Impact of ammonia emissions reduction on fine particulate matter: scenario analyses based on spatio-temporal kriging and random forest*

2017–2020 **BSc Economics and Statistics (L33)**, *University of Turin*, Italy
110/110 cum laude. Thesis: *The importance of confidence in economics and its possible assessment through Big Data*

Participation to Summer Schools:

2024 **Computer Vision for Earth Observation**, *IEEE IADF GRSS*, Benevento, Italy
This school focused on applying Computer Vision (CV) methods to address challenges in remote sensing. It consisted of a series of lectures on the existing methods utilized for analyzing satellite images, along with the challenges encountered. Each lecture was followed by a practical session where we practiced the techniques discussed in the lecture using Python.

2023 **Random Forests: Basics, Extensions and Applications**, *ECAS-SFdS*, Fréjus, France
In this course, I learned the basics of tree-based methods and popular tree ensemble methods. Furthermore, several extensions of random forests were presented with applications coming from several domains. Finally, attention was given to the relationship between random forests and other common machine-learning techniques.

Research Activity

Fields of interest:
Deep learning, Spatio-Temporal Data, Foundation Model, Earth System Modeling with Machine Learning (experience with weather forecasting, water resources and air pollution), Earth Observation, Physics-Guided ML, and eXplainable AI (XAI) for spatio-temporal models.

Scientific Publications:

[1] Ferraris, S., Meo, R., Pinardi, S., Salis, M., and Sartor, G. Machine learning as a strategic tool for helping cocoa farmers in Côte d'Ivoire. *Sensors*, 23(17):7632, 2023. `https://doi.org/10.3390/s23177632`.

[2] Fusta Moro, A., Salis, M., Zucchi, A., Cameletti, M., Golini, N., and Ignaccolo, R. Ammonia emissions and

fine particulate matter: evidence in Lombardy. In *ASA 2022 Data-Driven Decision Making - Book of Short Papers. ISBN: 979-12-215-0106-3*. 2023. https://doi.org/10.36253/979-12-215-0106-3.40.

[3] Salis, M., Sartor, G., Pellegrino, M., Ferraris, S., Atto, A. M., and Meo, R. What's behind this water table depth forecasting? RISE application for spatial, temporal, and spatio-temporal explanations. In *Proceedings of AI4IA 2024 - AI4CC: International Workshop on Artificial Intelligence for Climate Change. vol. 3883*. 2024. https://ceur-ws.org/Vol-3883/paper2_AI4CC5.pdf.

[4] Salis, M., Atto, A. M., Ferraris, S., and Meo, R. Time distributed deep learning models for purely exogenous forecasting: Application to water table depth prediction using weather image time series. *Environmental Modelling and Software*, 193:106568, 2025. https://doi.org/10.1016/j.envsoft.2025.106568.

[5] Salis, M., Sartor, G., Pellegrino, M., Meo, R., Ferraris, S., and Atto, A. Extending model-agnostic xai methods for regression tasks in spatio-temporal domains. In S. Džeroski, J. Levatić, G. Pio, and N. Simidjievski, editors, *Discovery Science*, pages 301–315. Springer Nature Switzerland, 2025. ISBN 978-3-032-05461-6. https://doi.org/10.1007/978-3-032-05461-6_20.

[6] Sartor, G., Salis, M., Pinardi, S., Saracik, O., and Meo, R. Deep learning tools to support deforestation monitoring in the Ivory Coast using SAR and optical satellite imagery. *International Journal of Applied Earth Observation and Geoinformation*, 144:104849, 2025. https://doi.org/10.1016/j.jag.2025.104849.

[7] Salis, M., Sartor, G., Meo, R., Ferraris, S., and Atto, A. M. Pure and physics-guided deep learning approaches for spatio-temporal groundwater level prediction. *Machine Learning: Science and Technology*, 7(5):055016, 2026. doi:10.1088/2632-2153/ae9af0.

Conferences and Workshop Contributions:

12/2026 (planned) **REO2 Workshop, NeurIPS 2026, Paris**, Presentation of the accepted peer-reviewed, non-archival, short paper: Salis, M., Ibrahimi, S., Wesselkamp, M., Shams Eddin M.H., Almikaeel, W., Hertel, V., Mallick, S., Luise, I., Kuehnert, J., Karlbauer, M., Hickman, S., Xhonneux, S., and Lessig, C. Land and Atmospheric Forecasting with a Pretrained JEPA-based Earth System Model.

Awards

2024 Best AI4CC 2024 Paper Award, for the paper: What's Behind This Water Table Depth Forecasting? RISE Application for Spatial, Temporal, and Spatio-Temporal Explanations

2019 1st place, "Doing Economics Data Competition" (CORE ECON). Report: *The environmental costs of being selfish* available at: https://www.core-econ.org/the-winners-of-the-core-doing-economics-competition-2019/

Projects

09/2025 - 09/2026 **CINECA - ISCRA Project Class C**, *Deep Learning for Spatio Temporal Phenomena, Application to Environmental Data.*

Role: Principal Investigator. The project provides access to HPC (LEONARDO) resources for 80000 standard hours. https://www.hpc.cineca.it/hpc-access/access-cineca-resources/iscra-projects/

03/2026 **Horizon Europe**, *Weather Generator*, the goal of the project is to build an open, kilometer-scale foundation model of the coupled Earth system using state-of-the-art AI and high-performance computing.

Role: I am a researcher representing Météo France, a partner of the project focused on high-resolution weather forecasting. https://weathergenerator.eu/

Teaching

2023–2025 **High School Summer School (I–III ed.)**, *AI and Data Science applied to water studies*, Torino, Italy

These summer schools have been designed for high school students to introduce basic concepts of data science and machine learning, with some applications on water resources data using a low-code programming framework (KNIME).

2025 **Python Programming with AI tools**, *ISITP Innocent Manzetti*, Aosta, Italy

This project was developed with the ISITP Innocent Manzetti (Aosta, IT) and consisted of 4 lectures (4 hours each) in which I taught high school students Python programming notions and how modern AI tools (chatbot) could be wisely used to enhance the quality and the speed of programming. `https://www.lavalleenotizie.it/scuola/limpatto-delluso-dellintelligenza-artificiale-progetto-al-manzetti-di-aosta/`

Reviewer activity

In Journals:

IEEE Transactions on Neural Networks and Learning Systems

In Conference and Workshop:

AI4IA 2024 - AI4CC: International Workshop on Artificial Intelligence for Climate Change: `https://ai4cc-2024.unibs.it/home-page`

AI 4 Science 2025 - AI 4 Space track: `https://ai4science.si/ai-space/`

AI4IA 2026 - AI4CC: International Workshop on Artificial Intelligence for Climate Change: `https://sites.google.com/unibs.it/ai4cc2026`

Thesis Supervision

2025 Marco Pellegrino, MSc thesis: *Explaining Groundwater Level Predictions: a RISE-Based Approach on Images Time-Series*.

2025 Marco Luddeni, MSc thesis: *AI in Education: Using Generative AI to Support Programming*.

2025 Dennis Montesi, MSc thesis: *Spatio-Temporal Graph Neural Networks for Groundwater Level Forecasting*.

Languages

Italian Native

English Academic

French B1 (Basic)